

Nitrogen efficiencies in a national nitrogen budget perspective

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KEY CLAIM

National nitrogen budgets allow benchmarking of a society's nitrogen application

BRIEF JUSTIFICATION, REMAINING CHALLENGES

Tracing the Nitrogen cascade is limited by many factors, specifically by data limitation. Using elemental balances on multiple scales provides a way to consolidate available information and to validate observations and trends (Winiwarter et al., 2025a, 2025b). Detailed guidance available also assures consistency of data compilation by differing institutions (Schäppi et al., 2025). This is the basis for a roll-out of the methodology currently performed by the Long-Range Transboundary Air Pollution Convention operating under the UN Economic Commission for Europe.

National nitrogen budgets are based on robust statistical data (available on a country scale) and provide input to numerous indicators to trace nitrogen pollution and impacts, also allowing to assess potential intervention points.

While benchmarking over time (comparison of developments of key parameters and indicators over time) is straightforward and can likewise be transferred to spatial comparison between countries after normalization (e.g. using flows per person, per area or per GDP), economic structures and regional particularities may strongly affect results, also in terms of resulting efficiencies. Comparisons between countries always need to integrate considerations such as whether a country uses N in relatively inefficient sectors such as agriculture only, or if also industrial use (with inherently higher efficiencies) is also to be considered.

Further data collection and evaluation, also on different scales, will be needed to understand the full potential of this tool. Reflecting physical flows allows to more closely connect between sources of pollution and their impact. Nitrogen-related impacts are always linked to the respective substrates, and hence their timing and specific location are highly relevant. A country scale may be too coarse a resolution to provide the desired connection, nevertheless they can support assessment of indicators or of footprint tools that fully lack such spatio-temporal aspects.

POTENTIAL FOR SYNERGY WITH OTHER CONTRIBUTIONS

National nitrogen budgets allow to connect N flows between different environmental pools. They provide a uniform framework to collect all information required for multiple indicators (NUE, N balance, N impacts).

- N footprint approaches may harvest data that otherwise are difficult to obtain
- by comparing different countries, the potentially diverse impacts due to similar N quantities can be tested and hence uncertainty derived
- National nitrogen budgets may benefit from a focus on issues identified in footprints (such as following up from country differences identified in footprints)

QUESTIONS TO SEED THE DISCUSSION

1. Which spatial scales are most suitable for relating N impacts, and how can national nitrogen budgets be rescaled?
2. Which parameters would nitrogen footprint approaches wish to extract from national N budgets?

REFERENCES

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